

# Probability Primer

## CS60050: Machine Learning

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Jul 23 and 24, 2026

# Agenda

To brush up basics of probability and random variables.

# Resources

- § “Probability, Statistics, and Random Processes for Electrical Engineering”, 3rd Edition, Alberto Leon-Garcia - [PSRPEE] - Alberto Leon-Garcia
- § “Machine Learning: A Probabilistic Perspective”, Kevin P. Murphy - [MLAPP] - Kevin Murphy:

# Introduction

- § Probability theory is the study of uncertainty.
- § The mathematical treatise of probability is very sophisticated, and delves into a branch of analysis known as **measure theory**.
- § We, however, will go through only basics of probability theory at a level appropriate for our course.

# Introduction

- § Probability is the Mathematical language for quantifying *uncertainty*. The starting point is to specify random experiments, sample space and set of outcomes.
- § A **random experiment** is an experiment in which the outcome varies in an unpredictable fashion when the experiment is repeated under the same conditions.
- § An **outcome** is a result of the random experiment and it can not be decomposed in terms of other results. The **sample space** of a random experiment is defined as the set of all possible outcomes. An outcome and the sample space of a random experiment will be denoted as  $\zeta$  and  $S$  respectively.

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# Introduction

## § Examples of random experiment

- ▶ Flipping a coin
- ▶ Rolling a die
- ▶ Flipping a coin twice
- ▶ Pick a number  $X$  at random between zero and one, then pick a number  $Y$  at random between zero and  $X$ .

- ▶  $S_1 = \{H, T\}$
- ▶  $S_2 = \{1, 2, 3, 4, 5, 6\}$
- ▶  $S_3 = \{HH, HT, TH, TT\}$
- ▶  $S_4 = \{(x, y) : 0 \leq y \leq x \leq 1\}$ .

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§ The corresponding sample spaces will be

- ▶  $S_1 = \{H, T\}$
- ▶  $S_2 = \{1, 2, 3, 4, 5, 6\}$
- ▶  $S_3 = \{HH, HT, TH, TT\}$
- ▶  $S_4 = \{(x, y) : 0 \leq y \leq x \leq 1\}.$

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- ▶ Getting even number when rolling a die,  $E_2 = \{2, 4, 6\}$
- ▶ Number of heads equal to number of tails when flipping a coin twice,  $E_3 = \{HT, TH\}$
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§ Three events are of special importance.

- ▶ **Simple event** are the outcomes of random experiments.
- ▶ **Sure event** is the sample space  $S$  which consists of all outcomes and hence always occurs.
- ▶ **Impossible or null event**  $\phi$  which contains no outcomes and hence never occurs.

# Introduction

§ **Set of events** (or **event space**)  $\mathcal{F}$ : A set whose elements are subsets of the sample space (*i.e.*, events).  $\mathcal{F} = \{A : A \subseteq S\}$ .  $\mathcal{F}$  is really a “set of sets”.

§  $\mathcal{F}$  should satisfy the following three properties.

- ▶  $\phi \in \mathcal{F}$
- ▶  $A \in \mathcal{F} \implies A^c (\triangleq S \setminus A) \in \mathcal{F}$
- ▶  $A_1, A_2, \dots \in \mathcal{F} \implies \cup_i A_i \in \mathcal{F}$

# Introduction

- § Probabilities are numbers assigned to events of  $\mathcal{F}$  that indicate how “likely” it is that the events will occur when a random experiment is performed.
- § Let a random experiment has sample space  $S$  and event space  $\mathcal{F}$ . Probability of an event  $A$  is a function  $P : \mathcal{F} \rightarrow \mathbb{R}$  that satisfies the following properties
  - ▶  $P(A) \geq 0, \forall A \in \mathcal{F}$
  - ▶  $P(S) = 1$
  - ▶ If  $A_1, A_2, \dots \in \mathcal{F}$  are disjoint events (i.e.,  $A_i \cap A_j = \emptyset$  for  $i \neq j$ ) then,  
 $P(\cup_i A_i) = \sum_i P(A_i)$
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# Introduction

## § Properties

- ▶  $P(A^c) = 1 - P(A)$
- ▶  $P(A) \leq 1$
- ▶  $P(\phi) = 0$
- ▶ If  $A \subseteq B$ , then  $P(A) \leq P(B)$ .
- ▶  $P(A \cap B) \leq \min(P(A), P(B))$
- ▶  $P(A \cup B) \leq P(A) + P(B)$

# Conditional Probability

§ **Conditional probability** provides whether two events are related in the sense that knowledge about the occurrence of one, say  $B$ , alters the likelihood of occurrence of the other say,  $A$ .

§ This conditional probability is defined as,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\begin{aligned} P(A|B) &= P(A) \\ \frac{P(A \cap B)}{P(B)} &= P(A) \\ P(A \cap B) &= P(A)P(B) \end{aligned}$$

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§ Two events  $A$  and  $B$  are **independent** (denoted as  $A \perp B$ ) if the knowledge of occurrence of one does not change the likelihood of occurrence of the other. This translates to the condition for independence as,

$$\begin{aligned} P(A|B) &= P(A) \\ \frac{P(A \cap B)}{P(B)} &= P(A) \\ P(A \cap B) &= P(A)P(B) \end{aligned}$$

# Total Probability Theorem

§ Let  $B_1, B_2, \dots, B_n$  be exhaustive and mutually exclusive events such that each of these events has positive probabilities. Then for any event  $A$ , the *total probability theorem* says,

$$P(A) = \sum_{i=1}^n P(A|B_i)P(B_i) \quad (1)$$

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**§ Proof:** Since,  $B_1, B_2, \dots, B_n$  are exhaustive (*i.e.*, their union covers the whole sample space),  $A = (A \cap B_1) \cup (A \cap B_2) \cup \dots (A \cap B_n)$

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$$\begin{aligned} P(A) &= P((A \cap B_1) \cup (A \cap B_2) \cup \dots (A \cap B_n)) \\ &= P(A \cap B_1) + P(A \cap B_2) + \dots + P(A \cap B_n) \\ &\quad (\text{as } B_i\text{'s are mutually exclusive}) \\ &= \sum_{i=1}^n P(A \cap B_i) = \sum_{i=1}^n P(A|B_i)P(B_i) \end{aligned}$$

# Total Probability Theorem

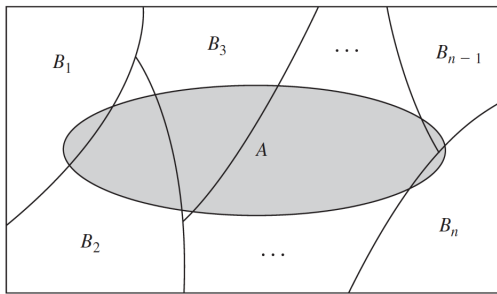


Figure credit: [PSRPEE] - Alberto Leon-Garcia

- § This is also known as **marginalization** operation.
- § Such exhaustive and mutually exclusive events  $B_1, B_2, \dots, B_n$  are also said to form a **partition** of the sample space.

# Bayes Rule

- § The total probability theorem is often used in conjunction with the Bayes' Rule that relates conditional probabilities of the form  $P(B|A)$  with conditional probabilities of the form  $P(A|B)$ .
- § Let the events  $B_1, B_2, \dots, B_n$  partitions a sample space such that each of the  $P(B_i)$ 's are non-negative. The Bayes' rule states,

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{P(A)} = \frac{P(A|B_i)P(B_i)}{\sum_{i=1}^n P(A|B_i)P(B_i)} \quad (2)$$

- § Bayes' rule is a very important tool for inference in machine learning.  $A$  can be thought of as the “effect” and  $B_i$ 's are several “causes” that can result in the effect. From the probabilities of the causes ( $B_i$ 's) resulting in the effect ( $A$ ) and the probability of the causes ( $B_i$ 's) to occur frequently, the probability that a particular cause ( $B_i$ ) is the reason behind the effect ( $A$ ) is computed.

# Random Variables

- § Statistics and Machine Learning are concerned with data. The link to sample space and events to data is **Random Variables**.
- § A random variable is a mapping  $(X : S \rightarrow \mathbb{R})$  from the sample space to real values that assigns a real number  $(X(\zeta))$  to each outcome  $(\zeta)$  in the sample space of a random experiment.

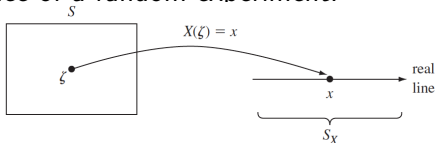


FIGURE 3.1

A random variable assigns a number  $X(\zeta)$  to each outcome  $\zeta$  in the sample space  $S$  of a random experiment.

Figure credit: [PSRPEE] - Alberto Leon-Garcia

- § We will use the following notation: capital letters denote random variables, e.g.,  $X$  or  $Y$ , and lower case letters denote possible values of the random variables, e.g.,  $x$  or  $y$ .

# Random Variables

## § An example from [PSRPEE] - Alberto Leon-Garcia

### Example 3.1 Coin Tosses

A coin is tossed three times and the sequence of heads and tails is noted. The sample space for this experiment is  $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$ . Let  $X$  be the number of heads in the three tosses.  $X$  assigns each outcome  $\zeta$  in  $S$  a number from the set  $S_X = \{0, 1, 2, 3\}$ . The table below lists the eight outcomes of  $S$  and the corresponding values of  $X$ .

|            |     |     |     |     |     |     |     |     |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|
| $\xi$ :    | HHH | HHT | HTH | THH | HTT | THT | TTH | TTT |
| $X(\xi)$ : | 3   | 2   | 2   | 2   | 1   | 1   | 1   | 0   |

$X$  is then a random variable taking on values in the set  $S_X = \{0, 1, 2, 3\}$ .

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§ Since the value of a random variable is determined by the outcome of the experiment, we may assign probabilities to the possible values of the random variable.

$$P(X = x) = P(\{\zeta \in S; X(\zeta) = x\}) \quad (3)$$

# Random Variables

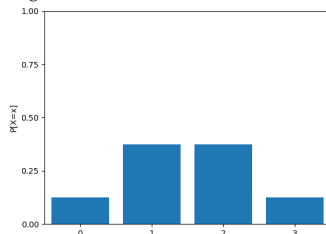
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$$P[X = 0] = P[\{TTT\}] = \frac{1}{8}$$

$$P[X = 1] = P[\{HTT, THT, TTH\}] = P[\{HTT\}] + P[\{THT\}] + P[\{TTH\}] = \frac{3}{8}$$

$$P[X = 2] = P[\{HHT, HTH, THH\}] = P[\{HHT\}] + P[\{HTH\}] + P[\{THH\}] = \frac{3}{8}$$

$$P[X = 3] = P[\{HHH\}] = \frac{1}{8}$$



Plot generated by: `discrete_prob_dist_plot`

from [MLAPP] - Kevin Murphy

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# Discrete Random Variables and PMF

- § A **discrete random variable**  $X$  is defined as a random variable that can take at most a countable number of possible values, *i.e.*,  
 $S_X = \{x_1, x_2, x_3, \dots\}$ .
- § A discrete random variable is said to be **finite** if its range is finite, *i.e.*,  $S_X = \{x_1, x_2, x_3, \dots, x_n\}$ .
- § The probabilities of events involving a discrete random variable  $X$  forms the **Probability Mass Function (PMF)** of  $X$  and it is defined as (ref eqn. (3)),

$$P_X(x) = P(X = x) = P(\{\zeta \in S; X(\zeta) = x\} \text{ for real } x) \quad (4)$$

- § Note that  $P_X(x)$  is a function of  $x$  over the real line, and that  $P_X(x)$  can be nonzero only at the values  $\{x_1, x_2, x_3, \dots\}$

# Continuous Random Variables and PDF

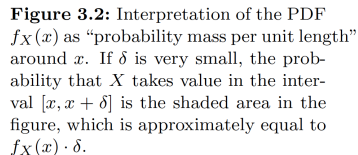
- § Random variables with a continuous range of possible experimental values are quite common.
- §  $X$  is a **continuous random variable** if there exists a non-negative function  $f_X(x)$ , defined for all real  $x \in (-\infty, \infty)$ , having the property that for any set  $B$  of real numbers,  $P(X \in B) = \int_B f_X(x)dx$ . The function  $f_X(x)$  is called the **probability density function (PDF)** of the random variable  $X$ .
- § Some properties of PDFs
  - ▶  $P(-\infty < X < \infty) = \int_{-\infty}^{\infty} f_X(x)dx = 1$
  - ▶  $P(a \leq X \leq b) = \int_a^b f_X(x)dx$
  - ▶ If we let  $a = b$  in the preceding, then  $P(X = a) = \int_a^a f_X(x)dx = 0$
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The diagram consists of two parts. On the left, a large gray oval is labeled "Sample Space". Inside it is a smaller white oval labeled "Event  $\{a \leq X \leq b\}$ ". An arrow points from this event oval to the right. On the right, a graph of a probability density function (PDF) is shown. The vertical axis is labeled "PDF  $f_X(x)$ " and the horizontal axis is labeled " $x$ ". The curve represents the PDF. The area under the curve between two points  $a$  and  $b$  on the  $x$ -axis is shaded with diagonal lines. An arrow from the event oval in the sample space points to this shaded area, indicating that the probability of the event is the integral of the PDF over that interval.

Fig credit: MIT Course: 6.041-6.43, Lecture Notes



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## Cumulative Distribution Function

- § We have defined PMF and PDF for discrete and continuous random variables respectively.
- § **Cumulative Distribution Function (CDF)** is a concept that is applicable to both discrete and continuous random variables. It is defined as,

$$F_X(x) = P(X \leq x) = \begin{cases} \sum_{k \leq x} P_X(k) & X : \text{discrete} \\ \int_{-\infty}^x f_X(t) dt & X : \text{continuous} \end{cases} \quad (5)$$

## Cumulative Distribution Function

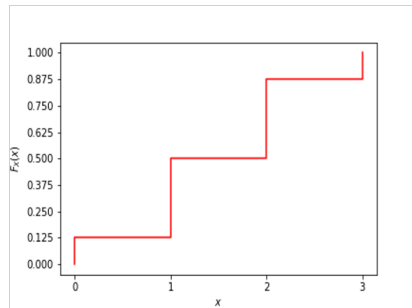
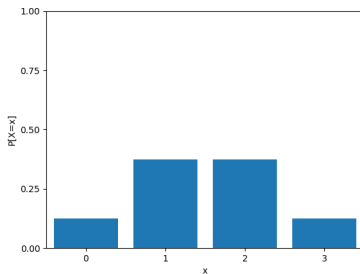
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- § For continuous random variables, the cumulative distribution function  $F_X(x)$  is differentiable everywhere. Naturally, in these cases, PDF is the derivative of the CDF.

$$f_X(x) = \frac{dF_X(x)}{dx}$$

## Cumulative Distribution Function



# Cumulative Distribution Function

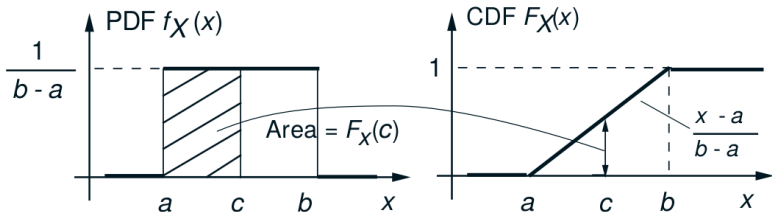
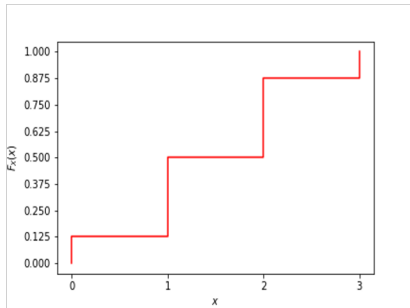
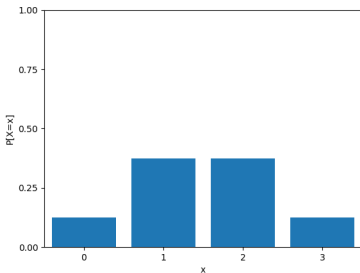


Fig credit: MIT Course: 6.041-6.43, Lecture Notes

# CDF - Some Properties

$$\S \quad 0 \leq F_X(x) \leq 1$$

$$\S \quad \lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\S \quad \lim_{x \rightarrow \infty} F_X(x) = 1$$

$$\S \quad x \leq y \implies F_X(x) \leq F_X(y)$$

# Expectation

§ The **expected value/expectation/mean** of a random variable is defined as:

$$\mathbb{E}[X] = \begin{cases} \sum_x x P_X(x) & \text{when } X \text{ is discrete} \\ \int x f_X(x) dx & \text{when } X \text{ is continuous} \end{cases} \quad (6)$$

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§ **Functions of random variable:** If  $Y = g(X)$  is a function of a random variable  $X$ , then  $Y$  is also a random variable, since it provides a numerical value for each possible outcome.

§ For a function of the random variable  $Y = g(X)$ , the expectation is, similarly, defined as,

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# Variance

- §  $\mathbb{E}[X]$  is also referred to as the **first moment** of  $X$ . Similarly the second moment is defined as  $\mathbb{E}[X^2]$  and in general, the  $n^{th}$  moment as  $\mathbb{E}[X^n]$
- § Another quantity of interest is the variance of a random variable  $x$ , denoted as  $\text{var}(X)$  and defined as  $\mathbb{E}[(X - \mathbb{E}[X])^2]$ . Variance provides a measure of dispersion of  $X$  around its mean  $\mathbb{E}[X]$ .
- § Another measure of dispersion is the standard deviation of  $X$ , which is defined as the square root of the variance  $\sigma_X = \sqrt{\text{var}(X)}$

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- § Another measure of dispersion is the standard deviation of  $X$ , which is defined as the square root of the variance  $\sigma_X = \sqrt{\text{var}(X)}$
- § Note that, using the rule for expected value of functions of random variables variance can be computed as,

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# Properties

## § Expectation

- ▶  $\mathbb{E}[a] = a$  for any constant  $a \in \mathbb{R}$
- ▶  $\mathbb{E}[af(X)] = a\mathbb{E}[f(X)]$  for any constant  $a \in \mathbb{R}$
- ▶  $\mathbb{E}[f(X) + g(X)] = \mathbb{E}[f(X)] + \mathbb{E}[g(X)]$

## § Variance

- ▶  $\text{var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - [\mathbb{E}[X]]^2$
- ▶  $\text{var}(a) = 0$  for any constant  $a \in \mathbb{R}$
- ▶  $\text{var}(af(X)) = a^2 \text{var}(f(X))$  for any constant  $a \in \mathbb{R}$

# Some Common Random Variables

## Discrete Random Variables

§ **Bernoulli** random variable: Takes two values 1 and 0 (or 'Head' and 'Tail'). The PMF is given by,

$$P_X(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases} \quad (9)$$

This is also written as  $P_X(x) = p^x(1 - p)^{1-x}$

§ It is used to model situations with just two random outcomes e.g., tossing a coin once.

§ For  $X \sim \text{Ber}(p)$ ,  $\mathbb{E}(X) = p$  and  $\text{var}(X) = p(1 - p)$ .

# Some Common Random Variables

## Discrete Random Variables

§ **Binomial** random variable: is used to model more complex situation e.g., the number of heads if a coin is tossed  $n$  times. The PMF is given by,

$$P_X(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n. \quad (10)$$

§ For  $X \sim \text{Bin}(n, p)$ ,  $\mathbb{E}(X) = np$  and  $\text{var}(X) = np(1-p)$ .

# Some Common Random Variables

## Discrete Random Variables

§ **Poisson** random variable: models situations where the events occur “completely at random” in time or space. The random variable counts the number of occurrences of the event in a certain time period or in a certain region in space. The PMF is given by,

$$P_X(x) = P(X = x) = \frac{\lambda^x}{x!} e^{-\lambda}, \quad x = 0, 1, 2, \dots \quad (11)$$

where  $\lambda$  is the average number of occurrences of the event in that specified time interval or region in space.

§ For  $X \sim \text{Poisson}(\lambda)$ ,  $\mathbb{E}(X) = \lambda$  and  $\text{var}(X) = \lambda$ .

# Some Common Random Variables

## Continuous Random Variables

§ **Uniform** random variable:  $X$  is a uniform random variable on the interval  $(a, b)$  if its probability density function is given by,

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & \text{if } a \leq x \leq b \\ 0, & \text{otherwise} \end{cases} \quad (12)$$

§ For  $X \sim \text{Uniform}(a, b)$ ,  $\mathbb{E}(X) = \frac{a+b}{2}$  and  $\text{var}(X) = \frac{(b-a)^2}{12}$ .



Fig credit: MIT Course: 6.041-6.43, Lecture Notes

# Some Common Random Variables

## Continuous Random Variables

§ **Exponential** random variable:  $X$  is a exponential random variable if its probability density function is given by,

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{if } x \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad (13)$$

§ For  $X \sim \text{Exponential}(\lambda)$ ,  $\mathbb{E}(X) = \frac{1}{\lambda}$  and  $\text{var}(X) = \frac{1}{\lambda^2}$ .

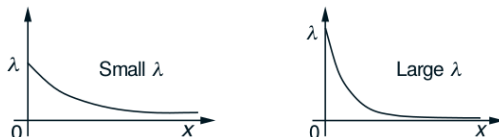


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# Some Common Random Variables

## Continuous Random Variables

§ **Gaussian/Normal** random variable:  $X$  is a Gaussian/Normal random variable if its probability density function is given by,

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (14)$$

§ For  $X \sim \text{Gaussian}(\mu, \sigma^2)$ ,  $\mathbb{E}(X) = \mu$  and  $\text{var}(X) = \sigma^2$ .

§ Gaussianity is Preserved by Linear Transformations. If  $X \sim \text{Gaussian}(\mu, \sigma^2)$  and if  $a, b$  are scalars, the random variable  $Y = aX + b$  is also Gaussian with mean and variance  $\mathbb{E}(Y) = a\mu + b$  and  $\text{var}(Y) = a^2\sigma^2$  respectively.

# Two Random Variables

§ Many random experiments involve several random variables. For example, temperature and pressure of a room during different days.

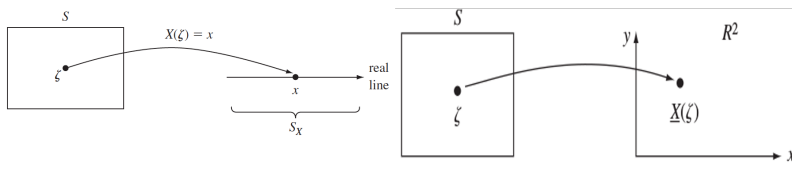


Figure credit: [PSRPEE] - Alberto Leon-Garcia

§ Consider two discrete random variables  $X$  and  $Y$  associated with the same experiment. We will use the notation  $P(X = x, Y = y)$  to denote  $P(X = x \text{ and } Y = y)$ .

# Two Random Variables

§ The **Joint PMF** of the two random variables  $X$  and  $Y$  is defined as,

$$\begin{aligned} P_{X,Y}(x, y) &= P(X = x, Y = y) \\ &= P(\{\zeta \in S; X(\zeta) = x, Y(\zeta) = y\} \text{ for real } x \text{ and } y) \end{aligned} \quad (15)$$

§  $P_X(x)$  and  $P_Y(y)$  are sometimes referred to as the **marginal PMFs**, to distinguish them from the joint PMF.

§ The marginal and the joint PMFs are related in the following way (ref eqn. (1), the total probability theorem),

$$P_X(x) = \sum_y P_{X,Y}(x, y) \text{ and } P_Y(y) = \sum_x P_{X,Y}(x, y) \quad (16)$$

# Two Random Variables

§ Similar to PDFs for single random variable, **joint PDF** for two continuous random variables is defined. for sets  $A$  and  $B$  of real numbers,

$$P(X \in A, Y \in B) = \int_B \int_A f_{X,Y}(x, y) dx dy \quad (17)$$

§ Similarly, **joint CDF** is also defined.

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y) = \begin{cases} \sum_{l \leq y} \sum_{k \leq x} P_{X,Y}(k, l) & X, Y : \text{discrete} \\ \int_{-\infty}^y \int_{-\infty}^x f_{X,Y}(u, v) du dv & X, Y : \text{continuous} \end{cases} \quad (18)$$

§ Differentiation for continuous random variables, yields

$$f_{X,Y}(x, y) = \frac{dF_{X,Y}(x, y)}{dy dx}$$

# Some Useful Relations

§ Marginal CDF can be obtained by setting the value of the other Random Variable to  $\infty$ , i.e.,  $F_X(x) = F_{X,Y}(x, \infty)$  and  $F_Y(y) = F_{X,Y}(\infty, y)$ .

§ Similar relations exist between marginal and joint PDFs.

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \text{ and } f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$$

§ Conditional PMF and Marginal PMF for discrete variables are related as,  $P_{Y|X}(y|x) = \frac{P_{X,Y}(x,y)}{P_X(x)}$  assuming that  $P_X(x) \neq 0$ .

§ Similar relation is there for continuous random variables.

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} \text{ provided } f_X(x) \neq 0.$$

# Joint Expectations

- § Similar expectation and moment rules exist for joint moments and expectation as in the case of a single random variable.
- § Considering  $Z = g(X, Y)$  as a function of two random variables, the expectation of  $Z$  can be found as,

$$\mathbb{E}[Z] = \begin{cases} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dx dy & X, Y \text{ continuous} \\ \sum_i \sum_j g(x_i, y_j) P_{X,Y}(x_i, y_j) & X, Y \text{ discrete} \end{cases} \quad (19)$$

- § Expectation of a sum of random variables is the sum of the expectations of the random variables.

$$\mathbb{E}[X_1 + X_2 + X_3 + \cdots] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \mathbb{E}[X_3] + \cdots \quad (20)$$

# Joint Moments, Correlation, and Covariance

§ The  $jk^{th}$  **joint moment** of  $X$  and  $Y$  is defined as,

$$\mathbb{E}[X^j Y^k] = \begin{cases} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^j y^k f_{X,Y}(x, y) dx dy & X, Y \text{ continuous} \\ \sum_m \sum_n x_m^j y_n^k P_{X,Y}(x_m, y_n) & X, Y \text{ discrete} \end{cases} \quad (21)$$

§ When  $j = k = 1$ , the corresponding moment  $\mathbb{E}[XY]$  gives the correlation between  $X$  and  $Y$ . If  $\mathbb{E}[XY] = 0$ ,  $X$  and  $Y$  are said to be **orthogonal**.

§ The  $jk^{th}$  **central moment** of  $X$  and  $Y$  is defined as

$$\mathbb{E}[(X - \mathbb{E}(X))^j (Y - \mathbb{E}(Y))^k]$$

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§ When  $j = k = 1$ , the corresponding central moment  $\mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))]$  is called the **covariance** between  $X$  and  $Y$ .

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# Joint Moments, Correlation, and Covariance

§ Covariance can also be expressed as  $\text{COV}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$

§ If  $X$  and  $Y$  are independent, then  $\text{COV}(X, Y) = 0$ , i.e.,  
 $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$

§ Correlation coefficient turns covariance into a normalized scale between  $-1$  to  $1$ .

$$\rho_{X,Y} = \frac{\text{COV}(X, Y)}{\sqrt{\text{VAR}(X)}\sqrt{\text{VAR}(Y)}} = \frac{\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]}{\sqrt{\text{VAR}(X)}\sqrt{\text{VAR}(Y)}} \quad (22)$$

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§ For example, let  $X \sim \mathcal{U}(-1, 1)$  and  $Y = X^2$ . Clearly,  $Y$  is dependent on  $X$ , but it can be shown that  $\rho_{X,Y} = 0$ .

$$\mathbb{E}[X] = \frac{-1 + 1}{2} = 0, \text{VAR}[X] = \frac{(1 - (-1))^2}{12} = \frac{1}{3}$$

$$\mathbb{E}[Y] = \mathbb{E}[X^2] = \text{VAR}[X] + (\mathbb{E}[X])^2 = \frac{1}{3} - 0^2 = \frac{1}{3}$$

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§ If  $X$  and  $Y$  are independent random variables, then random variables defined by any pair of functions  $g(X)$  and  $h(Y)$  are also independent, i.e., if  $P(XY) = P(X)P(Y)$  then  $P(g(X)h(Y)) = P(g(X))P(h(y))$ .

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# Conditional Expectation

§ The conditional expectation of  $Y$  given  $X = x$  is defined as,

$$\mathbb{E}[Y|x] = \int_{-\infty}^{\infty} y f_{Y|x}(y|x) dy \quad (24)$$

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§ The conditional expectation  $\mathbb{E}(Y|x)$  can be viewed as defining a function of  $x$ ,  $g(x) = \mathbb{E}(Y|x)$ . As  $x$ , is a result of a random experiment,  $\mathbb{E}(Y|x)$  is a random variable. So, we can find its expectation as,

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§ With some simple manipulation of the double integral it can be easily shown that  $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y|x]]$ . Sometimes, to remove confusion it is also written as  $\mathbb{E}_Y[Y] = \mathbb{E}_X[\mathbb{E}_Y[Y|x]]$  where the subscripts of the expectation sign denotes the expectation w.r.t. that random variable.

# Conditional Independence

§  $X$  and  $Y$  are **conditionally independent** given  $Z$  iff the conditional joint can be written as product of conditional marginals,

$$X \perp\!\!\!\perp Y|Z \Leftrightarrow P(X, Y|Z) = P(X|Z)P(Y|Z) \quad (26)$$

§ Conditional independence also implies,

$$X \perp\!\!\!\perp Y|Z \Rightarrow P(X|Y, Z) = P(X|Z) \text{ and } P(Y|X, Z) = P(Y|Z) \quad (27)$$

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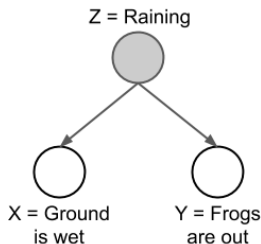
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# Multiple Random Variables

- § The notions and ideas can be generalized to more than two random variables. A **vector random variable**  $\mathbf{X}$  is a function that assigns a vector of real numbers to each outcome  $\zeta$  in the sample space  $S$  of a random experiment.
- § Uppercase boldface letters are generally used to denote vector random variables. By convention, it is a column vector. Each  $X_i$  can be thought of as a random variable itself.

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} = [X_1, X_2, \dots, X_n]^T$$

- § Possible values of the vector random variable are denoted by
- $$\mathbf{x} = [x_1, x_2, \dots, x_n]^T$$

# Multiple Random Variables

§ The **Joint PMF** of n-dimensional discrete random vector  $\mathbf{X}$

$$P_{\mathbf{X}}(\mathbf{x}) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) \quad (28)$$

§ Relation between the marginal and the joint PMFs,

$$P_{X_1}(x_1) = \sum_{x_2} \cdots \sum_{x_n} P_{\mathbf{X}}(\mathbf{x}) \quad (29)$$

§ Similarly, **joint CDF** is also defined.

$$\begin{aligned} F_{\mathbf{X}}(\mathbf{x}) &= P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n) \\ &= \begin{cases} \sum_{j \leq x_1} \sum_{k \leq x_2} \cdots \sum_{l \leq x_n} P_{\mathbf{X}}([x_1, x_2, \dots, x_n]^T) & \mathbf{X} : \text{discrete} \\ \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} \cdots \int_{-\infty}^{x_n} f_{\mathbf{X}}([u, v, \dots, w]^T) du dv \cdots dw & \mathbf{X} : \text{continuous} \end{cases} \end{aligned} \quad (30)$$

## Multiple Random Variables

§ The **joint PDF** of n-dimensional continuous random vector  $\mathbf{X}$

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{\partial^n F_{\mathbf{X}}(\mathbf{x})}{\partial x_1 \partial x_2 \cdots \partial x_3} \quad (31)$$

## § The marginal PDF

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{\mathbf{X}}([x_1, x_2, x_3, \cdots, x_n]^T) dx_2 dx_3 \cdots dx_n \quad (32)$$

## § The conditional PDF

$$f_{X_1/X_2, \dots, X_n}(x_1/x_2, \dots, x_n) = \frac{f_{\mathbf{X}}(\mathbf{x})}{f_{X_2, \dots, X_n}(x_2, \dots, x_n)} \quad (33)$$

## § Chain rule

$$\begin{aligned} f(x_1, x_2, \dots, x_n) &= f(x_n | x_1, \dots, x_{n-1}) f(x_1, \dots, x_{n-1}) \\ &= f(x_n | x_1, \dots, x_{n-1}) f(x_{n-1} | x_1, \dots, x_{n-2}) f(x_1, \dots, x_{n-2}) \\ &= f(x_1) \prod_{i=2}^n f(x_i | x_1, x_2, \dots, x_{i-1}) \end{aligned}$$

# Multiple Random Variables

§ There's also natural generalization of **independence**.

$$f(x_1, x_2, \dots, x_n) = f(x_1)f(x_2) \cdots f(x_n) \quad (35)$$

# Multiple Random Variables

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**§ Expectation:** Consider an arbitrary function  $g : \mathbb{R}^n \rightarrow \mathbb{R}$ . The expected value is,

$$\mathbb{E}[g(\mathbf{X})] = \int_{\mathbb{R}^n} g(\mathbf{X}) f_{\mathbf{X}}(\mathbf{x}) d\mathbf{x} \quad (36)$$

§ If  $g$  is a function from  $\mathbb{R}^n$  to  $\mathbb{R}^m$ , then the expected value of  $g$  is the element-wise expected values of the output vector, *i.e.*, if

$$\mathbf{g}(\mathbf{x}) = [g_1(\mathbf{x}), g_2(\mathbf{x}), \dots, g_n(\mathbf{x})]^T, \text{ then}$$

$$\mathbb{E}[\mathbf{g}(\mathbf{x})] = \left[ \mathbb{E}[g_1(\mathbf{x})], \mathbb{E}[g_2(\mathbf{x})], \dots, \mathbb{E}[g_n(\mathbf{x})] \right]^T$$

# Multiple Random Variables

§ **Covariance matrix:** For a random vector  $\mathbf{X} \in \mathbb{R}^n$ , covariance matrix  $\Sigma$  is  $n \times n$  square matrix whose entries are given by  $\Sigma_{ij} = \text{Cov}(X_i, X_j)$ .

$$\begin{aligned}
 \Sigma &= \begin{bmatrix} \text{Var}(X_1, X_1) & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_n) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2, X_2) & \cdots & \text{Cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \text{Cov}(X_n, X_2) & \cdots & \text{Var}(X_n, X_n) \end{bmatrix} \\
 &= \begin{bmatrix} \mathbb{E}[X_1^2] - \mathbb{E}[X_1]\mathbb{E}[X_1] & \cdots & \mathbb{E}[X_1 X_n] - \mathbb{E}[X_1]\mathbb{E}[X_n] \\ \mathbb{E}[X_2 X_1] - \mathbb{E}[X_2]\mathbb{E}[X_1] & \cdots & \mathbb{E}[X_2 X_n] - \mathbb{E}[X_2]\mathbb{E}[X_n] \\ \vdots & \ddots & \vdots \\ \mathbb{E}[X_n X_1] - \mathbb{E}[X_n]\mathbb{E}[X_1] & \cdots & \mathbb{E}[X_n^2] - \mathbb{E}[X_n]\mathbb{E}[X_n] \end{bmatrix} \\
 &= \begin{bmatrix} \mathbb{E}[X_1^2] & \cdots & \mathbb{E}[X_1 X_n] \\ \mathbb{E}[X_2 X_1] & \cdots & \mathbb{E}[X_2 X_n] \\ \vdots & \ddots & \vdots \\ \mathbb{E}[X_n X_1] & \cdots & \mathbb{E}[X_n^2] \end{bmatrix} - \begin{bmatrix} \mathbb{E}[X_1]\mathbb{E}[X_1] & \cdots & \mathbb{E}[X_1]\mathbb{E}[X_n] \\ \mathbb{E}[X_2]\mathbb{E}[X_1] & \cdots & \mathbb{E}[X_2]\mathbb{E}[X_n] \\ \vdots & \ddots & \vdots \\ \mathbb{E}[X_n]\mathbb{E}[X_1] & \cdots & \mathbb{E}[X_n]\mathbb{E}[X_n] \end{bmatrix} \\
 &= \mathbb{E}[\mathbf{X}\mathbf{X}^T] - \mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{X}^T] = \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^T]
 \end{aligned}$$

# Linear Transformations of Random Vectors

- § Suppose  $\mathbf{X}$  is some random vector and  $\mathbf{Y} = \mathbf{f}(\mathbf{X})$ , then we would like to know what are the first two moments of  $\mathbf{Y}$ .
- § Let  $\mathbf{f}(\cdot)$  is a linear function that is  $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$ , where  $\mathbf{X} \in \mathbb{R}^n$ ,  $\mathbf{A} \in \mathbb{R}^{m \times n}$ ,  $\mathbf{b} \in \mathbb{R}^m$  and  $\mathbf{Y} \in \mathbb{R}^m$ .
- § The mean will be  $\mathbb{E}[\mathbf{Y}] = \mathbb{E}[\mathbf{A}\mathbf{X} + \mathbf{b}] = \mathbf{A}\mathbb{E}[\mathbf{X}] + \mathbf{b}$ .
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$$\begin{aligned}\Sigma_{\mathbf{Y}} &= \mathbb{E}[(\mathbf{Y} - \mathbb{E}[\mathbf{Y}])(\mathbf{Y} - \mathbb{E}[\mathbf{Y}])^T] \\ &= \mathbb{E}[(\mathbf{A}\mathbf{X} + \mathbf{b} - \mathbf{A}\mathbb{E}[\mathbf{X}] - \mathbf{b})(\mathbf{A}\mathbf{X} + \mathbf{b} - \mathbf{A}\mathbb{E}[\mathbf{X}] - \mathbf{b})^T] \\ &= \mathbb{E}[\mathbf{A}(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^T \mathbf{A}^T] \\ &= \mathbf{A}\mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^T] \mathbf{A}^T = \mathbf{A}\Sigma_{\mathbf{X}}\mathbf{A}^T\end{aligned}\tag{38}$$

§ **Cross-covariance** between  $\mathbf{X}$  and  $\mathbf{Y}$  is  $\Sigma_{\mathbf{XY}} = \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{Y} - \mathbb{E}[\mathbf{Y}])^T]$

§ For  $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$ , it can be shown that  $\Sigma_{\mathbf{XY}} = \Sigma_{\mathbf{X}}\mathbf{A}^T$ .

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&= \mathbb{E}[\mathbf{A}(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^T \mathbf{A}^T] \\
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- § For  $\mathbf{Y} = \mathbf{AX} + \mathbf{b}$ , it can be shown that  $\Sigma_{\mathbf{XY}} = \Sigma_{\mathbf{X}}\mathbf{A}^T$ .